

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Pesticides - Set 1

Mark scheme

AQA spec 3.4.3.2.8; Pollution; textbook pp. 262-266

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for the correct answer.</i> • 25 g ha ⁻¹ (accept 24-27) (1)	1
01.2	<i>1 mark.</i> • The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species (1)	1
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • The MDAF (minimum dose always fatal) of the pest is about 50 g ha ⁻¹ (1) • This is higher than the MDNF of the bee (about 20 g ha ⁻¹) (1) • So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha ⁻¹ (1) • A dose of 30-40 g ha ⁻¹ would kill most pests with limited bee deaths (compromise) (1)	3
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Spray at night / when bees are not flying (1) • Spray when the crop is not in flower (1) • Spray on still days to reduce spray drift (1) • Use a systemic pesticide / seed coating rather than spraying (1) • Use smaller droplets for more even coverage (1)	2
01.5	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected (1) • A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) (1) • So fewer non-target deaths / less loss of predators, pollinators (1) • Fewer indirect (food chain) effects / fewer inter-species relationships lost (1) • Example: pyrethroids have high insect toxicity but low mammal toxicity (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	
02.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up (1) • High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) (1) • Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) (1) • Birds of prey are at the top of the food chain so receive the highest concentrations (1) • Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death (1)	4
02.2	<i>1 mark for each point.</i> • Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly (1) • Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens (1)	2
02.3	<i>1 mark.</i> • Pyrethroids are highly toxic to fish (even though mammal toxicity is low) (1)	1
02.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) (1) • They are moderately persistent so remain in the soil long enough to be absorbed (1) • Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats (1) • So bees / pollinators are less exposed than when the crop is sprayed (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
03.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $38.0 / 0.002$ (1) • $= 19\,000$ (19 000 times / 1.9×10^4) (1)	2
03.2	<i>1 mark.</i> • Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) (1)	1
03.3	<i>1 mark for each point.</i> • Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain (1) • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine (1) • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level (1) • Herons have long lives so accumulate over time (1)	4

Question	Marking guidance	Marks
03.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification (1) • They persist for decades so the damage continued long after use (1) • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify (1) • Benefit of preventing malaria deaths outweighs the risk (risk:benefit) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	
04.2	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Biological control (introduced predators / pathogens) (1) • Predator habitats (hedgerows, beetle banks) (1) • Crop rotation (1) • Companion / barrier crops (1) • Sterile male technique / pheromone traps (1) • Genetic resistance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
05.1	<i>1 mark.</i> • The dose that kills 50% of the (test) population (1)	1
05.2	<i>1 mark for each point.</i> • Insecticide X: LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well (1) • Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) (1) • Y has a much higher LD50 for the beetle (150) so few predators killed / more specific (1)	3
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • High water solubility so it is leached into rivers / groundwater / runoff (1) • Toxic to aquatic organisms downstream (1) • Mobile so it can affect areas away from the field (1) • Washed off the crop quickly so more frequent application needed (1)	2
05.4	<i>1 mark for each point.</i> • Long half-life = high persistence so it is not broken down quickly (1) • So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble) (1)	2
05.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Absorbed and translocated within the plant so protects new growth (1) • Not washed off by rain (1) • Does not need to cover all leaf surfaces / less spray needed (1) • Less spray drift so less harm to non-target species (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	
06.1	<i>1 mark.</i> • The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
06.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) • pH / presence of other chemicals may increase or decrease the rate of degradation (1) • Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
07.1	<i>1 mark for each point.</i> • Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) (1) • Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) (1)	2
07.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(200 - 25) / 200 \times 100$ (1) • = 87.5% (1)	2

Question	Marking guidance	Marks
07.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) (1) • Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths (1) • Higher doses caused sterility / death (1) • After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) (1) • Breeding success recovered so the population rose (1)	4
07.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • DDT persists in soil / sediments / food chains for many years so exposure continued (1) • Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly (1) • Other pressures (habitat loss, persecution) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	<i>Correct answer only.</i> • B - Low liposolubility and low persistence (1)	1
08.2	<i>1 mark for each point.</i> • Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown (1) • Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain (1)	2
08.3	<i>1 mark for each point.</i> • They are intended to be toxic (to the target species) (1) • They are deliberately released / spread into the environment (rather than being an accidental or waste release) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	